

CENG381 Stochastic Processes

Hitting Probabilities and Expected Hitting Times - Question Set 1

Notation used in this question set:

T_A	the first time the chain enters set A: $T_A = \min\{n \geq 0 : X_n \text{ is in } A\}$
$h(i, A)$	hitting probability: $P(\text{chain eventually enters } A \mid X_0 = i)$
$\tau(i, j)$	expected hitting time: $E[\text{first time to reach } j \mid \text{start at } i]$

Q1 (30 points). Consider the Markov chain with states {1, 2, 3, 4} and transition diagram:

```
1 --[1/2]--> 2,    1 --[1/2]--> 1 (self-loop)
2 --[1/2]--> 1,    2 --[1/2]--> 3
3 --[1/2]--> 2,    3 --[1/2]--> 4
4 --[1.0]--> 4 (absorbing)
```

- Write the transition matrix P for this chain.
 - Starting from state 2, what is the expected time to first reach state 4? Define $\tau(i) = E[\text{steps to reach 4} \mid X_0 = i]$ for $i = 1, 2, 3$. Note $\tau(4) = 0$. Set up the system of equations for $\tau(1)$, $\tau(2)$, $\tau(3)$ by conditioning on the first step.
 - Solve the system from part (b) and state $\tau(2)$.
-

Q2 (35 points). A fair coin is flipped repeatedly. Let X_n record the last symbol seen, tracking progress toward the pattern HTH (Heads-Tails-Heads). We model this with states representing how much of the target pattern has been matched so far:

```
State 0: no progress (starting state)
State 1: last flip was H (matched 'H')
State 2: last two flips were HT (matched 'HT')
State 3: last three flips were HTH (done - absorbing)
```

The transition probabilities (each flip is H with prob $1/2$, T with prob $1/2$) are:

```
From state 0: flip H -> go to state 1 (prob 1/2)
               flip T -> stay at state 0 (prob 1/2)
From state 1: flip T -> go to state 2 (prob 1/2)
               flip H -> stay at state 1 (prob 1/2)
From state 2: flip H -> go to state 3 (prob 1/2)
               flip T -> go to state 0 (prob 1/2)
From state 3: stay at state 3 (absorbing)
```

- Write the 4×4 transition matrix P .
- Let $\tau(i)$ = expected number of additional flips to reach state 3, starting from state i .

CENG381 Stochastic Processes

Hitting Probabilities and Expected Hitting Times - Question Set 1

Note $\tau(3) = 0$. Write the system of equations for $\tau(0)$, $\tau(1)$, $\tau(2)$.

c) Solve the system to find $\tau(0)$, the expected number of coin flips until the pattern HTH first appears.

Q3 (35 points). Consider the Coupon Collector problem: a cereal box contains one coupon from a set of n distinct coupons, chosen uniformly at random. A collector buys boxes until they have collected all n distinct coupons.

- a) Model this as a Markov chain where the state X_k = number of distinct coupons collected after k boxes. Write the transition probability $P(i, i+1)$ and $P(i, i)$ for $0 \leq i \leq n-1$.
- b) Let W_i = number of additional boxes needed to go from i distinct coupons to $i+1$ distinct coupons. Show that W_i follows a geometric distribution with success probability $(n-i)/n$, so that $E[W_i] = n/(n-i)$.
- c) Let T = total number of boxes needed to collect all n coupons. Show that $E[T] = n * (1 + 1/2 + 1/3 + \dots + 1/n)$. For $n = 5$, compute $E[T]$ numerically.